An update to this article is included at the end

Water Research 124 (2017) 472–481

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

The implications of drought and water conservation on the reuse of municipal wastewater: Recognizing impacts and identifying mitigation possibilities

Quynh K. Tran [a], David Jassby [a], Kurt A. Schwabe [b, c, *]

[a] Chemical and Environmental Engineering, University of California, Riverside, CA, USA
[b] School of Public Policy, University of California, Riverside, CA, USA
[c] Center for Global Food and Resources, University of Adelaide, Adelaide, Australia

ARTICLE INFO

Article history:
Received 13 May 2017
Received in revised form
3 July 2017
Accepted 26 July 2017
Available online 27 July 2017

Keywords:
Drought
Water conservation
Reuse
Treated municipal wastewater
Mitigation
Water quality

ABSTRACT

As water agencies continue to investigate opportunities to increase resilience and local water supply reliability in the face of drought and rising water scarcity, water conservation strategies and the reuse of treated municipal wastewater are garnering significant attention and adoption. Yet a simple water balance thought experiment illustrates that drought, and the conservation strategies that are often enacted in response to it, both likely limit the role reuse may play in improving local water supply reliability. For instance, as a particular drought progresses and agencies enact water conservation measures to cope with drought, influent flows likely decrease while influent pollution concentrations increase, particularly salinity, which adversely affects wastewater treatment plant (WWTP) costs and effluent quality and flow. Consequently, downstream uses of this effluent, whether to maintain streamflow and quality, groundwater recharge, or irrigation may be impacted. This is unfortunate since reuse is often heralded as a drought-proof mechanism to increase resilience. The objectives of this paper are two-fold. First, we illustrate—using a case study from Southern California during its most recent drought— how drought and water conservation strategies combine to reduce influent flow and quality and, subsequently, effluent flow and quality. Second, we use a recently developed regional water reuse decision support model (RWRM) to highlight cost-effective strategies that can be implemented to mitigate the impacts of drought on effluent water quality. While the solutions we identify cannot increase the flow of influent or effluent coming into or out of a treatment plant, they can improve the value of the remaining effluent in a cost-effective manner that takes into account the characteristics of its demand, whether it be for landscaping, golf courses, agricultural irrigation, or surface water augmentation.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

In response to water shortages and drought, municipalities are exploring a broad array of options to improve local water supply reliability and resilience. Demand management strategies that include both price (e.g., increasing block rate pricing) and nonprice approaches (e.g., low flush toilets, low flow showerheads) to reduce water consumption are one option (Baerenklau et al., 2014; Dupont, 2015). As Fig. 1 illustrates, demand management is likely a significant contributor to the noticeable reductions in per capita water use throughout the U.S. (MacKichan, 1957; MacKichan and Kammerer, 1961; Murray, 1968; Murray and Reeves, 1972, 1977; Solley et al., 1983, 1988, 1993, 1998; Hutson et al., 2004; Kenny et al., 2009; Maupin et al., 2014). Concurrently, and on the supply-side of the water scarcity ledger, agencies increasingly are augmenting local supplies through the use of treated municipal wastewater (Fig. 1) (MacKichan, 1957; MacKichan and Kammerer, 1961; Maupin et al., 2014; Murray, 1968; Murray and Reeves, 1972, 1977; Solley et al., 1983, 1988, 1993, 1998; Hutson et al., 2004; Groundwater Replenishment System (GWRS), 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015; IEUA, 2009, 2010, 2011, 2012, 2013a, 2013b, 2014, 2015a, 2015b, 2015c, 2016; Kenny et al.,

* Corresponding author. Present address: School of Public Policy, UC Riverside, Interdisciplinary South Building 4147, 3401 Watkins Dr., Riverside, CA, 92521, USA.
E-mail address: kurt.schwabe@ucr.edu (K.A. Schwabe).

http://dx.doi.org/10.1016/j.watres.2017.07.069
0043-1354/© 2017 Elsevier Ltd. All rights reserved.

Abbreviations

NPDES	National Pollutant Discharge Elimination System
IEUA	Inland Empire Utilities Agency
RWRM	Regional Water Reuse Decision-Support Model
GAMS	General Algebraic Modeling System
EC_w	Electrical Conductivity
TDS	Total Dissolved Solids
BOD	Biochemical Oxygen Demand
COD	Chemical Oxygen Demand
TOD	Total Oxygen Demands
TOC	Total Organic Carbon
TSS	Total Suspended Solids
RSC	Residual Sodium Carbonate
MF	Microfiltration
UF	Ultrafiltration
NF	Nanofiltration
RO	Reverse Osmosis
MGD	Million gallon per day
mg/L	milligram per liter
me/L	milliequivalent per liter
dS/m	DeciSiemens per meter
NTU	Nephelometric Turbidity Unit
MPN	Most Probable Number

2009; Schwabe and Connor, 2012; Bluefield Research., 2015; Tran et al., 2016). As an increasingly important element of a municipality's water portfolio, treated municipal wastewater can and often is used to irrigate local agriculture, golf courses, landscapes, and maintain local stream flows.

However, a simple water balance exercise reveals that demand management techniques and the use of treated municipal wastewater are linked, particularly through residential indoor water use (Min and Yeats, 2011; WaterSense, 2013). Consequently, if residential indoor water use decreases, the ability of treated municipal wastewater to help mitigate water shortages decreases while its cost likely increases. Specifically, indoor conservation can result in the generation of a more concentrated wastewater stream, with elevated concentrations of total dissolved solids (TDS), nitrogen species, and carbon (Stevens, 2015). This increase can present challenges to wastewater treatment plants, which were not designed to remove certain constituents (Sommer, 2014; Stevens, 2015; WaterWorld, 2015). Dissolved solids, in particular, represent a significant treatment challenge, as traditional wastewater treatment facilities are primarily concerned with nutrient removal. Therefore, conventional wastewater treatment facilities may have to invest significant resources to upgrade their treatment technologies, with the goal of complying with state and federal discharge limits, as well as to generate treated municipal wastewater that can be safely and reliably reused (U.S. EPA, 2004; Xia et al., 2004; García-Figueruelo et al., 2009; Norouzbahari et al., 2009; GWRS, 2014; IEUA, 2013a, 2013b; Sommer, 2014).

Without investments to address more concentrated wastewater streams, the water supplies of downstream communities might be adversely affected via multiple pathways. During extreme drought, natural flows in rivers decline, and wastewater discharges become a larger portion of surface water flow, resulting in a deterioration of surface water quality (Andersen et al., 2004; Whitehead et al., 2009; Benotti et al., 2010; U.S. Geological Survey (USGS), 2016a). The deterioration of water quality is illustrated by elevated concentrations of wastewater-derived contaminants, such as salts, nutrients, and/or other contaminants in the wastewater discharges, due to human and natural factors (e.g., population, water conservation, droughts, storms, etc.) (Whitehead et al., 2009; Benotti et al., 2010; USGS, 2013). An example of this phenomenon can be found in the Santa Ana River Basin, which is the largest stream system in Southern California (Bergamaschi et al., 2005; USGS, 2013). The flow in the Santa Ana River Basin is dominated by tertiary treated wastewater discharges generated by utilities throughout the basin (Bergamaschi et al., 2005). The water quality in the Santa Ana River Basin becomes poorer as the water flows downstream due to several factors, including recycling and reuse of wastewater within the basin (USGS, 2013). Consequently, any deterioration in treated wastewater effluent quality from upstream discharges has an immediate impact on downstream users.

The increase in wastewater contaminant concentrations may violate discharge limitations set by state and federal regulations (i.e., National Pollutant Discharge Elimination System (NPDES) permit) and result in penalties. Consider, again, the case of California in which violations result in mandatory minimum penalties from the Regional Water Quality Control Board (California State Water Resources Control Board, 2013). According to the Water Quality Enforcement Policy, California Water Code section 13,385(h) requires that a mandatory minimum penalties of $3000 be assessed by the Regional Water Quality Control Board for each serious violation. A serious violation is any discharge that exceeds the effluent limitation by at least 40% for Group I pollutants (Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Oxygen Demands (TOD), Total Organic Carbon (TOC), Total Suspended Solids (TSS), TDS, nutrients, detergents and oil, minerals, and metals), or by 20% for Group II pollutants (cyanide, total residual chlorine, all organic substances not listed under Group I).

Finally, poorer quality effluent can negatively impact sectors that rely on this water for irrigation, including golf courses, municipal and residential landscapes, and agriculture. Without mitigative actions within these sectors, plant growth and yields may suffer. Mitigative efforts to avoid such damages might include flushing with higher quality water, a response that can be equally costly and limited given freshwater supplies may be curtailed.

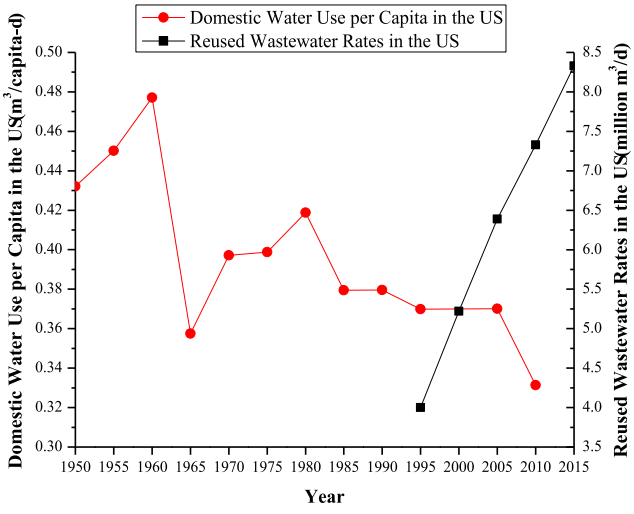


Fig. 1. Domestic water use per capita versus water reuse.

There have been several studies on the impacts of climate change on water/wastewater quality and availability (Andersen et al., 2004; Bergamaschi et al., 2005; Sprague, 2005; van Vliet and Zwolsman, 2008; Whitehead et al., 2009; Chen et al., 2009; Delpla et al., 2009; Benotti et al., 2010; Hrdinka et al., 2012; Jhansi and Mishra, 2013; Parks, 2016). To date, however, there have been few case studies investigating how drought may impact wastewater influent and effluent flow and quality, and how modernizing wastewater treatment trains can address these impacts. As the frequency and intensity of drought increases along with agency adoption of demand side management strategies, further understanding of how influent and effluent flows are impacted and cost-effective strategies to reduce such impacts is warranted.

The objective of this research is to investigate the impacts of drought and water conservation on the availability and quality of treated municipal wastewater. In particular, we investigate the impacts of drought on influent and effluent concentrations and identify cost-effective treatment responses to mitigate such impacts. After illustrating how the drought in California, including the state's response that mandated local water suppliers reduce water consumption by between 4 and 36%, impacted influent and effluent concentrations at a representative wastewater treatment plant in Southern California, we demonstrate how wastewater treatment trains can be upgraded to ensure continued high quality effluent discharges in the face of declining influent water quality.

2. Material and methods

2.1. Development of the regional water reuse decision-support model (RWRM) and optimization framework

The RWRM uses constrained linear programming methods to identify wastewater treatment trains optimized to deliver treated wastewater with specific water quality properties (Tran et al., 2016). Importantly, the programming model uses a constrained cost-minimization framework that identifies not only specific treatment processes, but also the relative volume fractions from individual parts of the treatment train that, when blended, provide for specific water quality attributes in the treated wastewater effluent. Constraints are imposed on various effluent water quality parameters to ensure that the specific water quality attributes meet regulatory and end-use demand parameters (e.g., threshold concentrations limits).

Seven feasible wastewater treatment technologies capable of providing a certain measure of salt removal were identified and arranged in eleven different wastewater treatment trains (Fig. 2): (1) activated sludge-granular filtration-MF-NF (TERT-MF-NF); (2) activated sludge-granular filtration-MF-RO (TERT-MF-RO); (3) activated sludge-granular filtration-UF-NF (TERT-UF-NF); (4) activated sludge-granular filtration-UF-RO (TERT-UF-RO); (5) activated sludge-granular filtration (TERT); (6) activated sludge-MF-nanofiltration (MF-NF); (7) activated sludge-microfiltration-RO (MF-RO); (8) activated sludge-UF-NF (UF-NF); (9) activated sludge-ultrafiltration-RO (UF-RO); (10) MBR-NF; and, (11) MBR-RO. Details on each treatment train is described in Supplementary Information. Critically, the model's flexibility allows for certain treatment steps to be bypassed in a given train in the interest of meeting particular effluent objectives/constraints at minimum cost. For example, if salt concentrations are not a concern, the model will not choose any desalination treatment technologies within a particular train given their relatively high cost.

The eleven treatment trains represent current and possible municipal wastewater treatment configurations, which could be implemented by Inland Empire Utilities Agency – Regional Water Recycling Plant #1 (IEUA – RP1) and many if not most other treatment plants throughout the U.S. to achieve the desired water quality. In these treatment trains, the removal of nutrients is achieved through the activated sludge or MBR processes, whereas MF and UF effectively remove large particles/molecules and bacteria/viruses and provide pre-treatment for the desalination steps. The desalination processes (NF and RO) effectively remove divalent and monovalent ions, respectively.

The cost associated with generating a unit volume of treated wastewater from each treatment train was estimated as the product of the unit treatment cost ($/m^3) for individual processes within a particular train multiplied by the blending fraction of the produced water associated with that process, as shown in equation (1):

$$C_T = \sum_{j=1}^{J_T} b_T(j)*c_T(j) \quad (1)$$

where C_T is the total treatment cost per m^3 of wastewater treated, T is the particular treatment train, j is the specific process within a treatment train T, which consists of j_T unique processes. The blending ratio of the produced water associated with process j for treatment train T is designated by $b_T(j)$, whereas the unit cost of each process is $c_T(j)$. The treatment cost is subject to two constraints. First, the total blending fractions of the produced water from each treatment train is restricted to one, corresponding to 100% of a given effluent volume as in equation (2), assuming no water loss in the treatment processes.

$$\sum_{j=1}^{J_T} b_T(j) = 1 \quad (2)$$

The concentration of different wastewater constituents (e.g., TDS, phosphorous) in the final blended product must be below specific threshold limits. In the current work, wastewater used for irrigation must comply with general irrigation guidelines established by the Food and Agriculture Organization of the United Nations (FAO) and the NPDES permit or, if discharged, must meet pre-drought treated effluent concentrations (which in our case study presented below is the year 2011) (Ayers and Westcot, 1985). Assuming water quality parameters are additive, the concentration of specific constituents can be calculated by:

$$\sum_{j=1}^{j_T} p_T(i,j)*b_T(j) \le a(i) \quad (3)$$

where, i is a specific wastewater constituent associated with process j within a given treatment train T, $p_T(i,j)$ is the concentration of specific constituent i resulting from process j, and $a(i)$ is the specific water quality threshold.

The optimization framework minimizes $C_T = \sum_{j=1}^{J_T} b_T(j)*c_T(j)$ subject to equations (2) and (3) using the General Algebraic Modeling System (GAMS) software. Thus, the RWRM is capable of producing a particular volume of effluent with specific water quality parameters at lowest cost by blending fractions of different treated wastewater effluents within a particular treatment train.

2.2. Modeling methodology

By modifying the RWRM's inputs and constraints, we applied the model to influent entering the IEUA – RP1 plant between 2011 and 2015, with the goal of identifying the lowest-cost solution to maintain pre-drought discharge quality, ensure treated effluent is suitable for reuse (irrigation), and protect downstream water

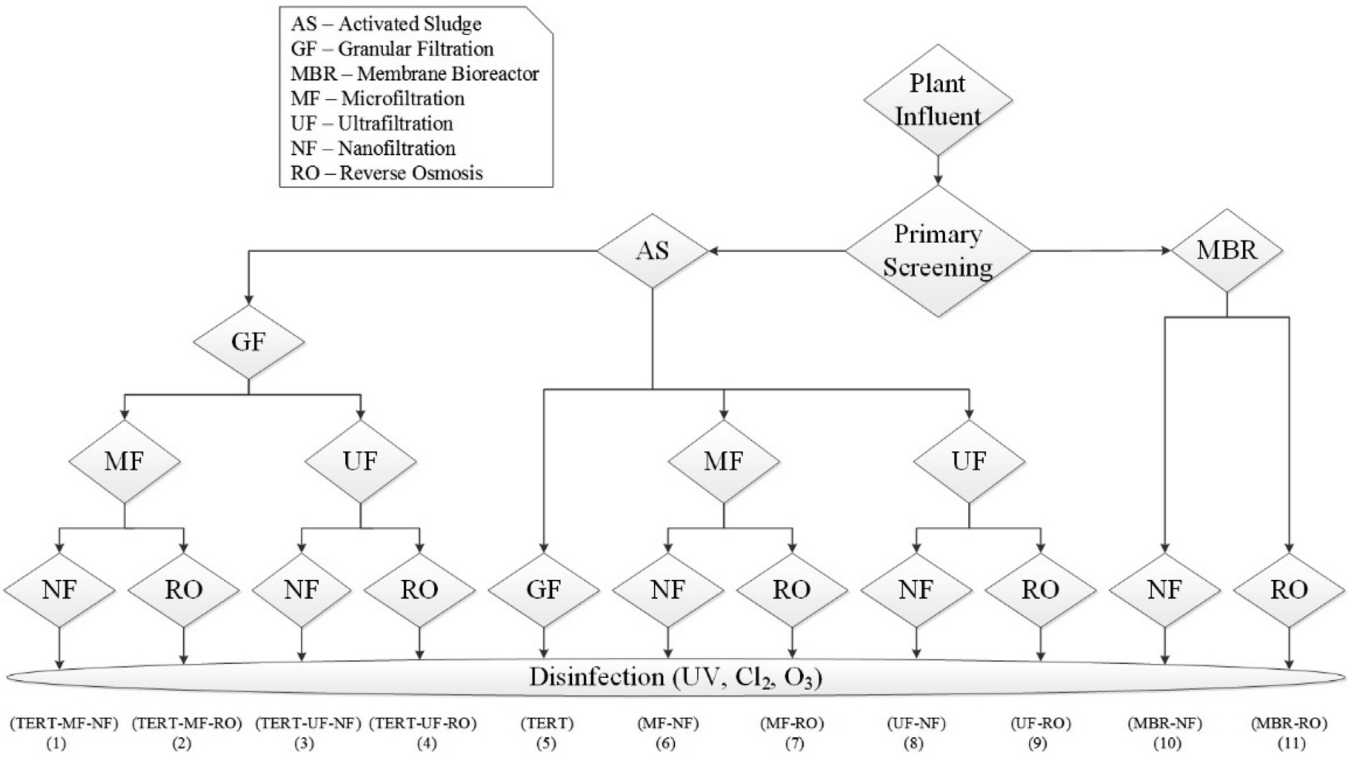


Fig. 2. Different treatment train processes corresponding to different GAMS logic.

quality. By identifying cost-effective blending combinations across different treatment processes (which could require infrastructure upgrades to the existing plant), the model suggests a method of restoring plant effluent quality to pre-drought conditions (i.e., 2011 effluent quality). In addition, the resultant water quality was evaluated for regulatory compliance (NPDES permit and Title 22 regulations) and safe agricultural practices based on general irrigation guidelines established by the FAO. Water quantity and quality data were collected from IEUA – RP1 to (a) investigate the changes in influent and effluent quality during drought and pre-drought periods (2011–2015) and (b) make recommendations on treatment train upgrades that would maintain pre-drought discharge quality for safe irrigation practices and state/federal compliance.

2.3. Model inputs

The RWRM inputs are comprised of four categories: the concentrations of water quality parameters in a given unit volume of influent wastewater, the removal performance associated with each treatment process, the concentration limits of water quality parameters, and the unit cost for each treatment process (Tables S1–S7). Conventional wastewater treatment processes consist of preliminary screening, primary, secondary, tertiary, and disinfection steps, which provide significant removals of nutrients and particles (Pescod, 1992). However, these conventional processes do not remove salinity (i.e., TDS) effectively (Pescod, 1992; USEPA, 2004). On the other hand, membrane separation processes, such as NF and RO, are commonly used in the water treatment industry to remove divalent and monovalent ions, respectively. The mechanisms responsible for ion removal are physical (size exclusion) and chemical (differences in diffusion rates) (Tchobanoglous et al., 2003; Crittenden et al., 2012). Therefore, to achieve a degree of desalination, these membranes must be incorporated into the wastewater treatment train.

IEUA – RP1's treatment train consists of screening, primary, secondary (activated sludge), tertiary (granular filtration), and chlorine disinfection (IEUA, 2013a, 2013b). According to California's Title 22 regulations, recycled water used for food crops, parks and playgrounds, school yards, residential landscaping, and unrestricted access golf courses shall be at least disinfected tertiary-recycled water, in which the turbidity does not exceed 2 NTU and the total coliform concentration must not exceed a 7-day median of 2.2 MPN/100 ml nor a 23 MPN/100 ml value in more than one sample in any 30-day period (California Department of Health Services, 2000). The plant's NPDES permit limits total nitrogen concentration in the water discharges to less than 10 mg/L, and limits ammonia-nitrogen (NH_4-N) to 4.5 mg/L to prevent excessive algae growth (Ayers and Westcot, 1985; California Department of Health Services, 2000; IEUA, 2015c). IEUA is in compliance with both NPDES and Title 22 regulations for the discharge and reuse of disinfected tertiary-recycled water (IEUA). Between 2011 and 2015, IEUA – RP1 effluent water quality was reported with NH_4-N concentrations of approximately 0.1 mg/L, which is far lower than the maximum concentration listed in the plant's permit (IEUA, 2015c; Price, 2016). Currently, there is no limitation on total phosphorus concentrations in IEUA – RP1 effluent. However, to prevent plant damage, phosphorous concentrations in irrigation water should be kept below 2 mg/L PO_4-P (Ayers and Westcot, 1985). In addition to phosphorous, elevated HCO_3^- concentrations exceeding 1.5 me/L (91.5 mg/L) in irrigation water can, over time, cause soil damage due to decreased soil permeability (Ayers and Westcot, 1985).

The 2011, 2015 IEUA – RP1 removal efficiencies of different wastewater constituents by different treatment steps are provided in the Supplementary Information (Tables S1 and S2). Wastewater quality parameters after each step of the IEUA – RP1 treatment process are listed in Tables S3 and S4; these data were obtained directly from the treatment plant for 2011 and 2015 (Price, 2016). Since IEUA – RP1 operates a conventional wastewater treatment

facility, data on membrane performance (i.e., % removal) were obtained from the literature (USEPA, 2004; Xia et al., 2004; García-Figueruelo et al., 2009; Norouzbahari et al., 2009). In instances where data were not available (italicized entries in Tables S1–S4), parameters were estimated based on typical removal percentages reported in the literature (e.g., activated sludge, microfiltration, ultrafiltration and granular filtration do not effectively remove salinity). Title 22 specifies a total coliform concentration of less than 2.2 MPN/100 ml for disinfected tertiary-recycled water; therefore, an additional disinfection step using either chlorine, ozone, or UV radiation must to be applied to the final treated effluent to comply with state guidelines.

The individual treatment costs associated with each process within a treatment train for large treatment facilities (>20,000 m^3/d or 5 MGD) are reported in Table S5 (Tran et al., 2016). Capital costs for each treatment step, including construction and administrative costs, were amortized over 15–20 years, depending on the process, plant capacities and time intervals. Operation and maintenance (O&M) costs cover the energy, chemicals, and maintenance costs (Leong et al., 2008). Data in Table S5 were adjusted to 2013 U.S. Dollars using the Engineering News Record Construction Cost Index and the Consumer Price Index (U.S. Bureau of Labor Statistics, 2000; Grogan, 2015). To summarize our general cost categories, currently it costs approximately $0.69 to produce a cubic meter of chlorine disinfected, tertiary-treated effluent at large conventional wastewater treatment facilities, such as IEUA – RP1; this cost consists of primary treatment ($0.02), activated sludge ($0.32), granular filtration ($0.33), and chlorine disinfection ($0.02) (Tran et al., 2016). In our analysis, sensitivity analysis over the range of interest rates (3%–10%) was explored and yielded negligible impacts on the magnitude of the amortized costs (Tables S8–S17).

3. Results and discussion

In this section, we present empirical evidence illustrating how drought and water conservation efforts by water agencies combine to reduce influent quality and flow thereby reducing the amount and quality of water available for reuse. We then identify, using the RWRM, a cost-effective treatment strategy that can mitigate the water quality impacts of drought and water conservation and return effluent water quality to levels consistent with pre-drought levels that also meet international irrigation recommendations. Finally, we relax the effluent quality constraints that were imposed to meet the pre-drought and international recommendations to illustrate how recognition of local conditions and end-use quality requirements can be used to further lower the treatment costs associated with drought response.

3.1. The role of drought and water conservation strategies on influent and effluent quality from 2011 to 2015

Changes in influent flow and quality arising from drought and water conservation can have deleterious effects on wastewater treatment plants and treated wastewater recipients. For IEUA – RP1, recipients of the disinfected, tertiary-treated effluent include industry, irrigators (e.g., for golf courses, schools, and parks), municipalities (e.g., for groundwater recharge), and the environment (e.g., discharge into a local creek) (IEUA, 2015a; 2016). As illustrated in Fig. 3, IEUA-RPI influent volumes dropped 14% (from 27.8 to 24 MGD) between 2011 and 2015 (Price, 2016). Such a decrease is not surprising given that in response to the drought that gripped California in 2013, its governor issued a Drought State of Emergency Proclamation in January, 2014, asking all Californians to reduce water consumption by 20% and directing local water suppliers to implement local water shortage contingency plans (CDWR, 2017).

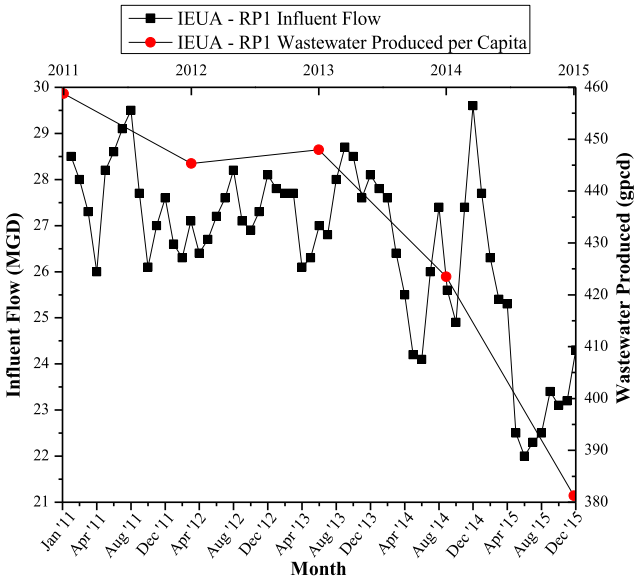


Fig. 3. IEUA – RP1 inflows vs. wastewater produced per capita in the IEUA – RP1 service area in 2011–2015.

As shown in Fig. 3, there is a significant drop in influent flow following the January 2014 proclamation. In April 2015, the State announced conservation mandates that were implemented in June 2015 and required water suppliers to meet conservation reduction targets that varied between 4 and 36% relative to 2013 use. IEUA's lowest inflow level immediately followed the 28% regional conservation mandate announcement it was expected to meet beginning in June 2015 (Arcadis U.S. Inc., 2016). From June to December 2015, IEUA was able to achieve monthly water use reductions of 27.5, 34, 28.5, 30.5, 26.5, 17.5, and 21%, respectively, relative to 2013.

While changes in influent flow mostly mirror changes in water conservation, other factors also matter. For instance, we speculate that California's storm of the decade in August 2014, and winter storms that occurred December 2014 to January 2015, and September to December 2015 (Sosnowski, 2014; Wiltgen and Breslin, 2014; Lloyd and Egger, 2015; Pamer et al., 2015; Pamer and Wolfe, 2015) increased influent flows via infiltration. Furthermore, freshwater usage within IEUA service area increased in July 2014 by 9% compared to the previous month which resulted in an 8% increase in inflow to the IEUA – RP1 plant (California State Water Resources Control Board, 2017). Consequently, freshwater usage and seasonal storms disrupted the influent wastewater correlation with conservation.

In addition to influencing the quantity of influent, drought and conservation may impact the quality of the influent via two mechanisms. First, less water use as a result of conservation activities such as fewer or shorter showers, fewer or fuller loads of laundry, or fewer flushes of the toilet can lead to an increase in the concentrations of the constituents in the generated wastewater. Second, the water supplies that agencies use to provide potable water can become degraded. As shown in Fig. 4, potable water within the IEUA service area, which is sourced from a mixture of ground and surface water supplies, exhibited a deterioration in quality over the 2011–2015 period in terms of TDS concentrations (City of Chino, 2010, 2011, 2012, 2013, 2014, 2015; City of Ontario, 2010, 2011; Cucamonga Valley Water District, 2010, 2011, 2012, 2013, 2014, 2015; Fontana Water Company, 2010, 2011, 2012, 2013, 2014, 2015; Ontario Municipal Utilities Company, 2012, 2013, 2014, 2015). From 2011 to 2015, average TDS concentrations in the potable water supply increased from 276 mg/L to 330 mg/L, a

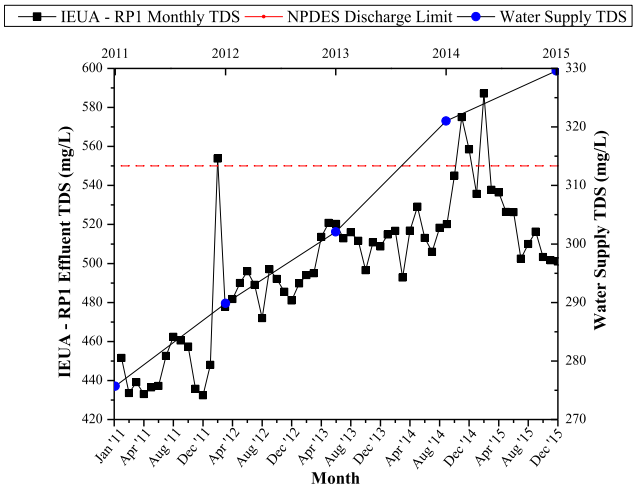


Fig. 4. TDS concentrations in 2010–2015.

20% increase. The increased TDS concentrations in potable water contributed to the increased TDS concentrations entering the IEUA – RP1 plant, which rose from 432 mg/L to 510 mg/L between 2011 and 2015.

The combination of poorer quality water supplies coupled with conservations activities resulted in an increase in pollution concentrations in the influent of IEUA – RP1 between 2011 and 2015 (Fig. 5). As the worst years of drought were between 2013 and 2015, the effects of the drought and subsequent indoor water conservation are clearly observed through an increase in the concentrations of certain constituents, including: TDS, electrical conductivity (EC_w), ions (sodium (Na^+), chloride (Cl^-), calcium (Ca^{2+}), HCO_3^-), and nutrients (Inset Fig. 5). In addition to the increase in TDS (18% increase), the concentrations of EC_w, Ca^{2+}, Na^+, Cl^-, and PO_4-P in the plant influent also showed an 8%, 13.6%, 16%, 6%, and 28% increase, respectively.

One consequence of higher constituent concentrations in the treatment plant influent is higher constituent concentrations in the effluent. As illustrated in Fig. 6, between 2011 and 2015 there was an 8–16% increase in many of the constituent concentrations at IEUA – RP1, including EC_w (8%) (Inset Fig. 6), TDS (7%), Ca^{2+} (16%), Na^+ (14%), HCO_3^- (11%), and Cl^- (13%). This is not surprising given that many standard wastewater treatment processes are not designed to remove these constituents effectively (USEPA, 2004). Another consequence is potential discharge violations and fines. Recalling Fig. 4, during the drought period, there were four discreet events in which IEUA – RP1 effluent discharges violated the NPDES discharge limitations in term of TDS concentrations. These violations can lead to financial penalties, complicated correction processes, and even permit termination (IEUA, 2015c; Price, 2016).

According to the NPDES permit, the monthly average concentration and loading rate for TDS shall not exceed 550 mg/L and 366,960 lb/day agency-wide, respectively. TDS concentrations exceeded these limits once in 2012, twice in 2014, and once in 2015. While concentrations exceeded permit limits throughout the drought, salt loadings were significantly below the NPDES limit highlighting the lower influent flows that give rise to higher constituent concentrations.

Overall, then, the drought and water conservation measures enacted because of the drought combined to both decrease the quantity and quality of wastewater inflow to IEUA – RP1. While conservation measures are critical to prolonging water supplies, indoor conservation measures may have adverse down-stream effects that reduce the role wastewater reuse may play in helping to mitigate the impacts of drought. Indeed, while average daily water reuse for irrigation volumes remained fairly steady within the IEUA service area from 2011 to 2015, the quality of the water declined. Furthermore, the lower overall treated volumes manifested themselves in lower discharges into surface waters by about 38%. While wastewater treatment plants are limited in mitigating the impacts of lower flows, there may be cost-effective opportunities to mitigate the increase in constituent concentrations that often accompany the lower flows, which we turn to next.

3.2. *Cost-effective treatment to mitigate poorer quality influent during drought*

What the above section has shown, then, is that at a time when

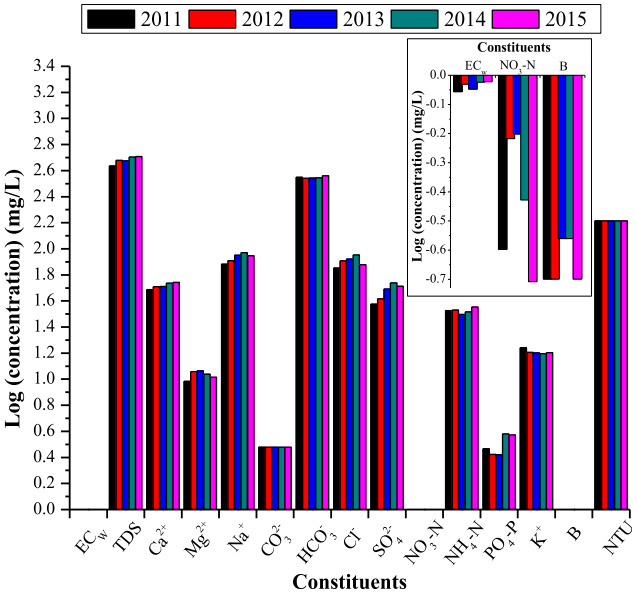


Fig. 5. IEUA – RP1 influent quality with EC_w, NO_3-N, and B concentrations (inset) in 2011–2015.

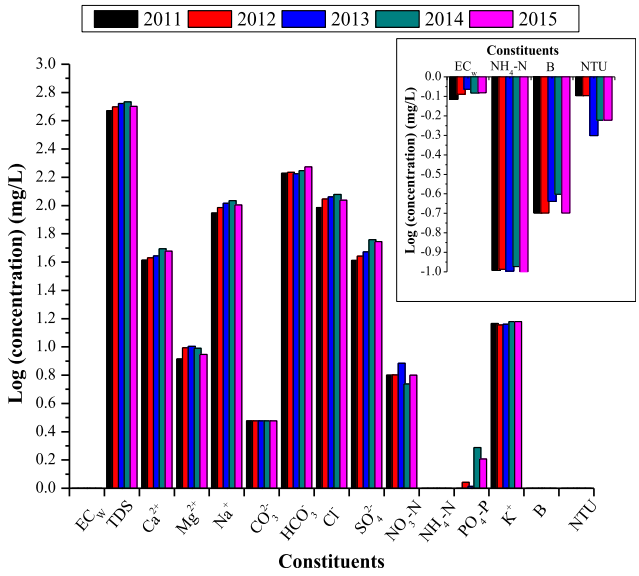


Fig. 6. IEUA – RP1 effluent quality with EC_w, NH_4-N, B, and turbidity (inset) in 2011–2015.

the reuse of treated municipal wastewater may be most useful—during drought—the quantity and quality of water available for reuse may decline, thereby reducing its ability to help mitigate the impacts of drought. This section investigates the ability of wastewater treatment plants to address one of the issues—reduced influent quality. Specifically, we illustrate how our model can be used to identify a cost-effective treatment strategy that plants may adopt to mitigate the impacts of drought on the quality of treated municipal wastewater.

To illustrate how wastewater treatment plants might cost-effectively mitigate the poorer quality influent during times of drought, we use the actual water quality parameters IEUA – RP1 confronted in 2015 as a measure of the poorer quality influent and effluent that may result during drought. We then investigate and apply a cost-effective treatment strategy that achieves comparable water quality parameters representative of what might be observed in a non-drought year in which the effluent is suitable for crop irrigation and can be safely used for downstream users without deteriorating surface or ground water quality. For our case here, we choose the water quality parameters observed in the effluent of the IEUA – RP1 plant in 2011 as an example of such a situation. As shown in Table 1, the concentrations of many water constituents in 2015 IEUA – RP1 effluent exceeded the levels observed in 2011, including the "no restriction" levels listed in Table S7. In particular, water quality parameters associated with salinity were sufficiently high in 2015 to warrant some additional treatment, including EC_w (0.828 vs. 0.7 dS/m), TDS (504 vs. 450 mg/L), Na^+ (4.42 vs. 3 me/L), NO_3-N (6.314 vs. 5 mg/L), Cl^- (3.08 vs. 3 me/L) and HCO_3^- (3.08 vs. 1.5 me/L).

To improve 2015 effluent quality to 2011 levels, as well as to be in compliance with FAO guidelines and NPDES limits, it is necessary for the wastewater treatment plant to implement some degree of desalination to reduce salt concentrations in the plant effluent. Traditional wastewater treatment processes, like the IEUA – RP1 plant, were not designed to reduce salinity; consequently, the salinity of some wastewater effluent used for irrigation is above the maximum tolerance for certain crops, which can harm both the soil and plants (Ayers and Westcot, 1985; Carrow et al., 2008; Peacock et al., 2012; Tran et al., 2016). The FAO general irrigation water quality guidelines specify that the use of irrigation water with water quality parameters that exceed certain minimums should be "slightly to moderately" restricted, depending on the concentration (Table S7) (Ayers and Westcot, 1985). So while we constrain our model to meet the 2011 IEUA – RP1 effluent quality levels, our model also requires that both the FAO guidelines and NPDES limits be met. These additional requirements that go above and beyond the 2011 water quality parameters affect the EC_w, TDS, Na^+, HCO_3^-, Cl^-, and NH_4-N, as noted by the asterisk in Table 1.

Using the RWRM, we identify the most cost-effective wastewater treatment trains that would improve wastewater effluent quality to sufficient quality under three different scenarios. First, our (baseline) Scenario A involves achieving the water quality standards achieved in 2011 at IEUA – RP1 with the additional FAO and NPDES requirements. The results, shown in the bottom row of Table 1, identifies the most cost-effective solution to consist of blending effluent treated by MF with effluent treated under the MF-RO train. Specifically, the cost-effective solution consists of blending a mix of 47.1% MF effluent with 52.9% MF-RO effluent, which then undergoes disinfection. Because the model has to choose a treatment train for which no constituent concentrations exceed those concentrations in 2011 or that meet FAO guidelines, we see that the binding constituent is HCO_3^-, with lower concentrations for all the other constituents. For example, TDS concentrations were 44% lower than TDS values measured in 2011 (253.45 mg/L vs. 450 mg/L).

In this sense, the model chooses a solution that overtreats the influent in every dimension except HCO_3^- and thus produces a superior quality effluent. The bottom row of Table 1 illustrates that the cost associated with this treatment scenario is $\$0.775/m^3$, which is 12% higher than the current treatment cost incurred by the IEUA – RP1 plant.

The second scenario—scenario B—imposes a less restrictive constraint on HCO_3^- concentrations, which were the binding constraint from the baseline scenario. The initial HCO_3^- constraint was based on FAO recommendations. HCO_3^- limits, as stated in the FAO's guidelines, are important because elevated HCO_3^-

Table 1
Baseline, drought, and scenario concentrations and costs.

	2015 IEUA – RP1 effluent quality (mg/l)	2011 IEUA – RP1 effluent quality, FAO guidelines, and NPDES discharge permit (mg/l)	Percent change from 2011 Values		
			Scenario A	Scenario B	Scenario C
Parameters					
EC (dS/m)	0.828	0.7[a]	−32%	−22%	−19%
TDS	504	450[a]	−44%	−35%	−33%
Ca^{2+}	47.58	41.16	−36%	−26%	−23%
Mg^{2+}	8.833	8.228	−40%	−30%	−27%
Na^+	101.08	69[a]	−38%	−28%	−25%
CO_3^{2-}	3	3	−52%	−45%	−42%
HCO_3^-	187.88	91.5[a]	0%	+16%	+20%
Cl^-	109.25	106.5[a]	−66%	−60%	−59%
SO_4^{2-}	55.5	40.92	−40%	−30%	−27%
NO_3-N	6.314	5[a]	−38%	−28%	−26%
NH_4-N	0.1	4.5[b]	−99%	−99%	−99%
PO_4-P	1.611	0.861	−14%	0%	+4%
K^+	15.08	14.64	−47%	−38%	−36%
B	0.2	0.2	−50%	−43%	−41%
TC (MPN/100 ml)	<2	<2	−100%	−100%	−100%
Turbidity (NTU)	0.6	0.8	−92%	−91%	−91%
% Blending					
MFCl	N/A	N/A	47.1%	55.0%	57.2%
RO-M	N/A	N/A	52.9%	45.0%	42.8%
Treatment cost ($/m^3)[c]	$0.69	$0.69	$0.775	$0.747	$0.740

[a] FAO guidelines.
[b] NPDES discharge limits.
[c] Costs based on 2013 U.S. Dollar.

concentrations can have negative effects on soil permeability, which impacts water infiltration particularly during periods of low humidity and high evaporation (Ayers and Westcot, 1985). Carbonate species will combine with Ca^{2+} and Magnesium (Mg^{2+}) and precipitate as Ca-$MgCO_3$. This process removes Ca^{2+} and Mg^{2+} from the soil water and results in an increase in the proportion of Na^{+} which, in turn, increases the sodium hazard in the soil (Ayers and Westcot, 1985). The effect of soil permeability is evaluated by residual sodium carbonate (RSC), which is an index of the sodicity hazard associated with the irrigation water. Water containing less than 1.25 me/L of RSC is considered safe for irrigation (Cockerham and Leinauer, 2011). The RSC is defined by the formula:

$$RSC\left(\frac{me}{L}\right) = \left(CO_3^{2-} + HCO_3^{-}\right) - \left(Ca^{2+} + Mg^{2+}\right)$$

Elevated concentrations of certain irrigation water constituents (10–20% greater than FAO guidelines) can be tolerated when considered with other factors, such as sodium adsorption ratio (SAR), RSC, and EC_w, that affect crop yield (Ayers and Westcot, 1985). Thus, relaxing HCO_3^- concentration constraints (scenario B) may be tolerable when HCO_3^- concentrations in irrigation water do not exceed 1.8 me/L or 109.8 mg/L (20% over the FAO limit), which corresponds to an RSC value of −0.84 me/L. The RSC value generated under scenario A was −0.18 me/L, which is far below the FAO's guidelines, again indicating *overtreatment* of the wastewater.

As shown in Table 1, the cost-effective strategy arising from relaxing the HCO_3^- constraint while still maintaining the 2011 concentration limits for the other constituents (scenario B) consists of a blend that is 55% MF effluent and 45% MF-RO effluent, again followed by chlorine disinfection. By relaxing the HCO_3^- constraint, the cost-effective solution relies less on the costlier MF-RO treatment than the strategy under scenario A, thereby lowering treatment costs by approximately 4%. Of course, costs are still higher compared to current 2015 costs, around 8%, but overall water quality improves for all of the remaining parameters except phosphorous (PO_4-P), which is now the binding constraint.

The last scenario considers relaxing the phosphorus constraint, a constraint defined by its 2011 value. While phosphorous is one of the essential nutrients for plant growth, excessive phosphorus can interfere with iron and zinc uptake by plant roots (Keeley and Whitney, 1998). In addition, excess phosphorus from wastewater discharge and runoff can contribute to algal blooms and low dissolved oxygen levels in water bodies, which in turn can harm aquatic species (USGS, 2016a, b). Currently, there is no limitation on the discharged total phosphorus in wastewater effluent for streams in California (USEPA, 2012). Therefore, phosphorus concentrations in treated wastewater could be above the threshold over which it impacts crop yield (2 mg/L PO_4-P) (Ayers and Westcot, 1985). As Table 1 illustrates, PO_4-P concentration limits that achieve the 2011 levels are significantly less than what is identified by the FAO. The final scenario, then, identifies the cost-effective solution of treating the 2015 influent to 2011 effluent levels except that in addition to relaxing the HCO_3^- constraint as imposed in scenario B, the PO_4-P constraints are relaxed to FAO standards.

As the final column in Table 1 illustrates, relaxing both the HCO_3^- and PO_4-P relative to scenario A (the 2011 levels with strict FAO constraints on HCO_3^-) results in a further reduction in the use of RO for treatment. As shown, the proportion of influent treated by MF increased to over 57%, while the amount treated by RO dropped to 43%, for about a 5% reduction in costs relative to scenario A yet still a 7% increase from current treatment costs. While this treatment strategy leads to an increase in 15 of the 16 constituent concentrations relative to scenario A, the effluent would still be considered overtreated relative to the 2011 values.

4. Conclusions

In times of drought, which are expected to increase in both frequency and severity, the availability of freshwater resources decrease and its quality, particularly with respect to salinity, often declines. In response, water agencies are increasingly implementing conservation measures to reduce demand while investing in the reuse of treated municipal wastewater to augment supply, both with the intention to increase resilience. Our results show that both the water supply effects of drought *and* the conservation measures enacted in response to it combine to reduce the quantity and quality of influent available for treatment which, in turn, reduces the quantity and quality of treated municipal wastewater for reuse under conventional treatment processes. Our modeling results also illustrate that cost-effective blending strategies can be implemented to mitigate the water quality effects, increasing the value of the remaining effluent for reuse, whether it be for surface water augmentation, groundwater replenishment, or irrigation of crops, golf courses, or landscapes. We also highlight the benefits of treating the wastewater based on the characteristics of demand. In our case study, relaxing the constraints on both HCO_3^- and PO_4-P concentrations relative to the pre-drought levels lead to reduced treatment costs while still achieving an effluent quality superior to pre-drought levels for all the other water quality parameters.

From a management and policy perspective, three significant conclusions can be drawn from our analysis. First, municipalities, cities, and regions that rely on both conservation and reuse as a means to address drought and water scarcity need to recognize the potential dependence of the latter on the former in terms of its effect on the potential supply of treated municipal wastewater. To the extent possible, efforts to promote and advocate for outdoor water conservation rather than indoor conservation break this dependence. Second, drought and the conservation measures enacted in response can result in poorer quality water, particularly with respect to salinity. Given that conventional treatment processes are not designed to address these higher constituent loads, the value of the remaining effluent likely decreases relative to the downstream demands it serves. Consequently, recognizing this relationship should help the recipients of the treated municipal wastewater better plan for such outcomes and thus engage in cost-effective adaptation. Finally, while wastewater treatment plants themselves cannot mitigate the reduced flows that are the result of the drought and conservation measures, our modeling results illustrate that cost-effective treatment trains can be developed to mitigate the water quality effects of drought and conservation thereby increasing the value of the remaining effluent for reuse. We propose that by working together, recipients of treated municipal wastewater and the agencies themselves can identify cost-effective strategies in terms of the degree of treatment that provides the greatest benefit to society.

Our conclusions here are not simply fodder for the academic mill. In our particular case study, treated wastewater makes up a significant portion of flows in the Santa Ana watershed, as well as Southern California's waterways. It plays an even more significant role during drought conditions, when precipitation and snowmelt decrease. Thus, it is important to consider how deteriorating wastewater effluent quality, and in particular, elevated salinity levels, impacts downstream users. Our modeling results demonstrate that incorporating a desalination step into the wastewater treatment process can alleviate some of these downstream concerns at a cost that is within 8% (currently \$0.69/m^3 vs. \$0.74/m^3 under scenario C) of current treatment costs. The resulting effluents are composed of partially desalinated wastewater that is suitable for crop irrigation and stream augmentation at a quality that protects wastewater treatment agencies from discharge violations and

prevents surface water quality from further deterioration.

Identification of such low-cost wastewater treatment strategies should be useful to municipalities, both in California and globally, as they continue to strive to improve their resilience to drought via demand side management strategies that include reducing indoor water use and supply augmentation strategies that include wastewater reuse (Schwabe and Connor, 2012). The RWRM model used in this research can be easily adapted to other applications to identify low-cost strategies given its flexibility and replicability. The model requires unit cost and effectiveness parameters on commonly used treatment technologies, parameters that are regularly reported in the academic literature and/or industry reports. The parameters that represent regulatory, surface water, or crop threshold constraints on effluent quality also can readily attained from public documents. For instance, this model was previously used to identify the most cost-effective treatment solutions when the treated wastewater effluent is used for irrigating citrus and turfgrass in Southern California (Tran et al., 2016). Consequently, the RWRM is a flexible and easily adaptable model that can assist water managers in their efforts to develop water portfolios that cost-effectively and reliably respond to drought and increasing water scarcity worldwide.

Acknowledgements

This work was supported by the USDA/AFRI Grant program [Grant number 2015-68007-23191] and NIFA-BARD Program [Project No. NB-8307-14].

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.watres.2017.07.069.

References

Andersen, C.B., Lewis, G.P., Sargent, K.A., 2004. Influence of wastewater-treatment effluent on concentrations and fluxes of solutes in the Bush River, South Carolina, during extreme drought conditions. Environ. Geosci. 11 (1), 28–41.
Arcadis U.S. Inc, 2016. 2015 Urban Water Management Plan Final, p. 450. Los Angeles, CA.
Ayers, R.S., Westcot, D.W., 1985. Water Quality for Agriculture. Food and Agriculture Organization of the United Nations.
Baerenklau, K.A., Schwabe, K.A., Dinar, A., 2014. The residential water demand effect of increasing block rate water budgets. Land Econ. 90 (4), 683–699.
Benotti, M.J., Stanford, B.D., Snyder, S.A., 2010. Impact of drought on wastewater contaminants in an urban water supply. J. Environ. Qual. 39 (4).
Bergamaschi, B.A., Kalve, E., Guenther, L., Mendez, G.O., Belitz, K., 2005. An Assessment of Optical Properties of Dissolved Organic Material as Quantitative Source Indicators in the Santa Ana River Basin, Southern California, p. 38.
Bluefield Research, 2015. U.S. Municipal Wastewater & Reuse: Market Trends, Opportunities and Forecasts, 2015-2025, p. 140.
California Department of Health Services, 2000. Water Recycling Criteria, Sacramento, CA.
California Department of Water Resources (CDWR), 2017. Governor's Drought Declaration, California Department of Water Resources.
California State Water Resources Control Board, 2017. June 2014-January 2017 Urban Water Supplier Report Dataset, State of California, CA.
California State Water Resources Control Board, 2013. Legislative Mandated Reports - Water Code Section 13385(o) Enforcement Report California State Water Resources Control Board, Sacramento, CA.
Carrow, R.N., Duncan, R.R., Huck, M.T., 2008. Turfgrass and Landscape Irrigation Water Quality: Assessment and Management. CRC Press.
Chen, B., Nam, S.-N., Westerhoff, P.K., Krasner, S.W., Amy, G., 2009. Fate of effluent organic matter and DBP precursors in an effluent-dominated river: a case study of wastewater impact on downstream water quality. Water Res. 43 (6), 1755–1765.
City of Chino, 2010. 2010 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Chino, 2011. 2011 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Chino, 2012. 2012 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Chino, 2013. 2013 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Chino, 2014. 2014 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Chino, 2015. 2015 Consumer Confidence Report. City of Chino, Chino, CA, p. 2.
City of Ontario, 2010. 2010. Annual Water Quality Report. City of Ontario, Ontario, CA, pp. 18–20.
City of Ontario, 2011. 2011 Annual Water Quality Report. City of Ontario, Ontario, CA, pp. 18–20.
Cockerham, S.T., Leinauer, B., 2011. Turfgrass Water Conservation. University of California, Agriculture and Natural Resources..
Crittenden, J.C., Harza, M.W., Hand, D.W., Howe, K.J., 2012. MWH's Water Treatment: Principles and Design. Wiley.
Cucamonga Valley Water District, 2010. 2010 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Cucamonga Valley Water District, 2011. 2011 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Cucamonga Valley Water District, 2012. 2012 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Cucamonga Valley Water District, 2013. 2013 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Cucamonga Valley Water District, 2014. 2014 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Cucamonga Valley Water District, 2015. 2015 Water Quality Report. Cucamonga Valley Water District, Rancho Cucamonga, CA, p. 2.
Delpla, I., Jung, A.-V., Baures, E., Clement, M., Thomas, O., 2009. Impacts of climate change on surface water quality in relation to drinking water production. Environ. Int. 35 (8), 1225–1233.
Dupont, D., 2015. Water Conservation: Thinking beyond the Tap. Handbook of Water Economics, pp. 372–387.
Fontana Water Company, 2010. 2010 Annual Water Quality Report, p. 4. Fontana, CA.
Fontana Water Company, 2011. 2011 Annual Water Quality Report, p. 4. Fontana, CA.
Fontana Water Company, 2012. 2012 Annual Water Quality Report, p. 4. Fontana, CA.
Fontana Water Company, 2013. 2013 Annual Water Quality Report, p. 4. Fontana, CA.
Fontana Water Company, 2014. 2014 Annual Water Quality Report, p. 4. Fontana, CA.
Fontana Water Company, 2015. 2015 Annual Water Quality Report, p. 4. Fontana, CA.
García-Figueruelo, C., Bes-Piá, A., Mendoza-Roca, J.A., Lora-García, J., Cuartas-Uribe, B., 2009. Reverse osmosis of the retentate from the nanofiltration of secondary effluents. Desalination 240 (1–3), 274–279.
Grogan, T., 2015. Dramatic Drop in Fuel Prices Starts to Affect the Market, Engineering News Record.
Groundwater Replenishment System (GWRS), 2008. Ground Water Replenishment System Annual Report 2008, pp. 1–3.
Groundwater Replenishment System (GWRS), 2009. Ground Water Replenishment System Annual Report 2009, pp. 1–3.
Groundwater Replenishment System (GWRS), 2010. Ground Water Replenishment System Annual Report 2010, pp. 1–3.
Groundwater Replenishment System (GWRS), 2011. Ground Water Replenishment System Annual Report 2011, pp. 1–3.
Groundwater Replenishment System (GWRS), 2012. Ground Water Replenishment System Annual Report 2012, pp. 1–3.
Groundwater Replenishment System (GWRS), 2013. Ground Water Replenishment System Annual Report 2013, pp. 1–3.
Groundwater Replenishment System (GWRS), 2014. Ground Water Replenishment System Annual Report 2014, pp. 1–3.
Groundwater Replenishment System (GWRS), 2015. Ground Water Replenishment System Annual Report 2015, pp. 1–3.
Hrdinka, T., Novický, O., Hanslík, E., Rieder, M., 2012. Possible impacts of floods and droughts on water quality. J. Hydro-Environ. Res. 6 (2), 145–150.
Hutson, S.S., Barber, N.L., Kenny, J.F., Linsey, K.S., Lumia, D.S., Maupin, M.A., 2004. Estimated Use of Water in the United States in 2000, p. 52. Reston, VA.
Inland Empire Utilities Agency (IEUA), 2013a. Consolidated FY 13/14 Recycled Water Annual Report Final.
Inland Empire Utilities Agency (IEUA). Regional Water Recycling Plant #4, Inland Empire Utilities Agency.
Inland Empire Utilities Agency (IEUA), 2009. Fiscal Year 2008-09 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2010. Fiscal Year 2009-10 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2011. Fiscal Year 2010-11 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2012. Fiscal Year 2011-12 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2013b. Fiscal Year 2012-13 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2014. Fiscal Year 2013-14 Recycled Water Annual Report, p. 3.
Inland Empire Utilities Agency (IEUA), 2015a. 2015 Annual Report - Chino Basin Recycled Water Groundwater Recharge Program, p. 160. Chino, CA.
Inland Empire Utilities Agency (IEUA), 2015b. Fiscal Year 2014-15 Recycled Water Annual Report, p. 3.

Inland Empire Utilities Agency (IEUA), 2015c. Waste Discharge Requirements for Master Reclamation Permit for Inland Empire Utilities Agency, p. 103. Santa Ana Region.
Inland Empire Utilities Agency (IEUA), 2016. Fiscal Year 2015-16 Recycled Water Annual Report, p. 3.
Jhansi, S.C., Mishra, S.K., 2013. Wastewater treatment and reuse: sustainability options. Consilience J. Sustain. Dev. 10 (1), 1–15.
Keeley, S., Whitney, D., 1998. A Guide to Turfgrass Nutrient Recommendations. Kansas State University, p. 5.
Kenny, J.F., Barber, N.L., Hutson, S.S., Linsey, K.S., Lovelace, J.K., Maupin, M.A., 2009. Estimated Use of Water in the United States in 2005, p. 52. Reston, VA.
Leong, L.Y., Kuo, J., Tang, C.-C., 2008. Disinfection of Wastewater Effluent-comparison of Alternative Technologies. Water Environment Research Foundation.
Lloyd, J., Egger, C., 2015. Storm Unleashes Record Rainfall across Southern California. NBCUniversal Media. Universal City, CA.
MacKichan, K.A., 1957. Estimated Use of Water in the United States, 1955, p. 21. Washington, D.C.
MacKichan, K.A., Kammerer, J.C., 1961. Estimated Use of Water in the United States, 1960, p. 50. Washington, D.C.
Maupin, M.A., Kenny, J.F., Hutson, S.S., Lovelace, J.K., Barber, N.L., Linsey, K.S., 2014. Estimated Use of Water in the United States in 2010, p. 56. Reston, VA.
Min, K., Yeats, S.A., 2011. Water conservation efforts changing future wastewater treatment facility needs. Fla. Water Resour. J. 39–41.
Murray, C.R., 1968. Estimated Use of Water in the United States, 1965, p. 62. Washington, D.C.
Murray, C.R., Reeves, E.B., 1972. Estimated Use of Water in the United States in 1970, p. 46. Reston, VA.
Murray, C.R., Reeves, E.B., 1977. Estimated Use of Water in the United States in 1975, p. 46. Reston, VA.
Norouzbahari, S., Roostaazad, R., Hesampour, M., 2009. Crude oil desalter effluent treatment by a hybrid UF/RO membrane separation process. Desalination 238 (1–3), 174–182.
Ontario Municipal Utilities Company, 2012. 2012 Water Quality Report. Ontario Municipal Utilities Company, Ontario, CA, pp. 5–7.
Ontario Municipal Utilities Company, 2013. 2013 Water Quality Report. Ontario Municipal Utilities Company, Ontario, CA, pp. 5–7.
Ontario Municipal Utilities Company, 2014. 2014 Water Quality Report. Ontario Municipal Utilities Company, Ontario, CA, pp. 5–7.
Ontario Municipal Utilities Company, 2015. 2015 Water Quality Report. Ontario Municipal Utilities Company, Ontario, CA, pp. 5–7.
Pamer, M., McDade, M.B., Mester, M., 2015. Storm System Makes for White Christmas in SoCal Mountains. Winds Whip Region, KTLA, Los Angeles, CA.
Pamer, M., Wolfe, C., 2015. Storm Cells Drop Rain on L.a. Basin, Triggering Flood Advisory. KTLA, Los Angeles, CA.
Parks, J., 2016. Data inquiry for research purpose. In: Tran, Q. (Ed.), Encina Wastewater Authority, Carlsbad, California.
Peacock, C.H., Miller, G.L., Martin, M.C., 2012. Irrigation Water Quality Problems. North Caroline State University, North Carolina, p. 2.
Pescod, M.B., 1992. Wastewater Treatment and Use in Agriculture. Food and Agriculture Organization of the United Nations, Rome, Italy.
Price, C., 2016. PRR - Wastewater Quality from IEUA Wastewater Treatment Plant. In: Tran, Q. (Ed.), IEUA, Chino, CA.
Schwabe, K.A., Connor, J.D., 2012. Drought issues in semi-arid and arid environments. Choices 27 (3), 1–5.
Solley, W.B., Chase, E.B., Mann Iv, W.B., 1983. Estimated Use of Water in the United States in 1980.
Solley, W.B., Merk, C.F., Pierce, R.R., 1988. Estimated Use of Water in the United States in 1985.
Solley, W.B., Pierce, R.R., Perlman, H.A., 1993. Estimated Use of Water in the United States in 1990, p. 86. Washington, D.C.
Solley, W.B., Pierce, R.R., Perlman, H.A., 1998. Estimated Use of Water in the United States in 1995.
Sommer, L., 2014. Record Drought Could Hurt Water Quality. KQED, San Francisco, CA.
Sosnowski, A., 2014. Biggest Storm in Years for California Unleashes Flooding Rain, Mudslides, P. California Storm of the Decade 2014, AccuWeather, PA.
Sprague, L.A., 2005. Drought effects on water quality in the south platte River Basin, Colorado. JAWRA J. Am. Water Resour. Assoc. 41 (1), 11–24.
Stevens, M., 2015. Unintended Consequences of Conserving Water: Leaking Pipes, Less Revenue, Bad Odors. The Los Angeles Times, Los Angeles, CA.
Tchobanoglous, G., Burton, F.L., Stensel, H.D., Metcalf, Eddy, 2003. Wastewater Engineering: Treatment and Reuse. McGraw-Hill Education.
Tran, Q.K., Schwabe, K.A., Jassby, D., 2016. Wastewater reuse for agriculture: development of a regional water reuse decision-support model (RWRM) for cost-effective irrigation sources. Environ. Sci. Technol. 50 (17), 9390–9399.
U.S. Bureau of Labor Statistics, 2000. CPI - Consumer Price Index Historical_US 1967. U.S. Bureau of Labor Statistics.
U.S. Environmental Protection Agency (U.S. EPA), 2004. In: Guidelines for Water Reuse. Office of Wastewater Management: U.S. Environmental Protection Agency, Washington, DC.
U.S. Environmental Protection Agency (U.S. EPA), 2012. Action towards Limiting Total Nitrogen, Total Phosphorus, and Total Inorganic Nitrogen Loads from NPDES - Permitted Facilities. U.S. EPA.
U.S. Geological Survey, 2016a. California Drought. U.S. Geological Survey, California Water Science Center.
U.S. Geological Survey, 2013. Santa Ana Basin. U.S. Geological Survey, San Diego, CA.
U.S. Geological Survey, 2016b. Phosphorus and Water. USGS.
van Vliet, M.T.H., Zwolsman, J.J.G., 2008. Impact of summer droughts on the water quality of the Meuse river. J. Hydrol. 353 (1–2), 1–17.
WaterSense, 2013. In: Reduce Outdoor Water Use U.S. Environmental Protection Agency. U.S. EPA, Washington, D.C, p. 1.
WaterWorld, 2015. PPIC to Research Impact of Drought on Water Quality. WaterWorld, Tulsa, OK.
Whitehead, P., Wilby, R., Battarbee, R., Kernan, M., Wade, A.J., 2009. A review of the potential impacts of climate change on surface water quality. Hydrol. Sci. J. 54 (1), 101–123.
Wiltgen, N., Breslin, S., 2014. Southern California Flooding: Mudslides, Flash Flooding California Damage Homes, Leave at Least 1 Dead, the Weather Channel.
Xia, S., Nan, J., Liu, R., Li, G., 2004. Study of drinking water treatment by ultrafiltration of surface water and its application to China. Desalination 170 (1), 41–47.

Update

Water Research

Volume 125, Issue , 15 November 2017, Page 521

DOI: https://doi.org/10.1016/j.watres.2017.09.053

Water Research 125 (2017) 521

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

Corrigendum

Corrigendum to "The implications of drought and water conservation on the reuse of municipal wastewater: Recognizing impacts and identifying mitigation possibilities" [Water Res. 124 (2017), 472–481]

Quynh K. Tran [a], David Jassby [a], Kurt A. Schwabe [b, c, *]

[a] Chemical and Environmental Engineering, University of California, Riverside, CA, USA
[b] School of Public Policy, University of California, Riverside, CA, USA
[c] Center for Global Food and Resources, University of Adelaide, Adelaide, Australia

The authors regret to inform that there was a typographical error in the above-cited paper related to the units on the right-hand side axis on Fig. 3. The corrections are given below.

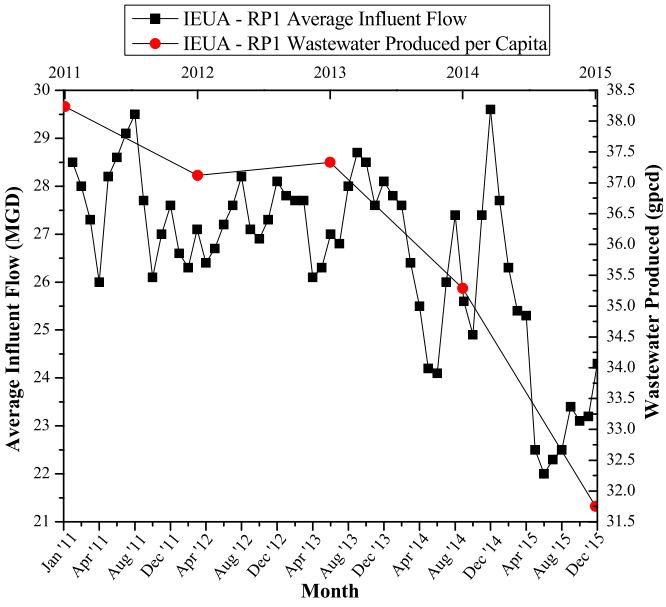


Fig. 3. IEUA – RP1 inflows vs. wastewater produced per capita in the IEUA – RP1 service area in 2011–2015.

Correcting the error of units on the right-hand axis does not change the conclusions or change any other numerical results of the paper. We thank Dr. Denny Parker from Brown and Caldwell for informing us of this issue.

DOI of original article: https://doi.org/10.1016/j.watres.2017.07.069.
* Corresponding author. School of Public Policy, University of California, Riverside, CA, USA.
E-mail address: kurt.schwabe@ucr.edu (K.A. Schwabe).

https://doi.org/10.1016/j.watres.2017.09.053
0043-1354/© 2017 Elsevier Ltd. All rights reserved.